

Predictive Modeling for Crop Recommendation Using Soil Nutrients and Climatic Parameters

Group Number: 8

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Goal of the Project

Crop selection has significant implications for yield, resource use and economic return, and is a key decision for farmers to make for each piece of land. This is based on personal experience and intuition which can sometimes be faulty as the soil type and climatic conditions change. The aim of this project is to create and test automatic crop recommendation system based on machine learning that can recommend the most appropriate crop for a particular location with the help of 7 measurable soil and climatic parameters like N, P, K, temperature, humidity, pH of soil, rainfall. We intend to use a labeled dataset with 22 different types of crop and train several classification models to evaluate their performance and to find a model that works well on unseen conditions and can be packaged as a lightweight prediction tool that can be used to aid agricultural decision making.

Materials and Method

Dataset

This project comes with the publicly available Crop Recommendation Dataset: <https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset>. The data set consists of 2200 data points for 22 classes of crops (100 data points per class, equally distributed) each one of which has 7 numeric features: N, P, K, temperature, humidity, pH, and rainfall with no missing features.

Code Repository

Colab file: <https://tinyurl.com/Crop-Recommendation-Training>

Data Preprocessing

- **Validation:** ensured accurate column data types, value was not null, and balance in classes for 22 crops.
- **train/test split:** A stratified split with random seed = 42 has been used for train/test split (80/20), which ensures the same class distribution in both splits.
- **Feature scaling:** All seven numeric features were standardized with the function StandardScaler from scikit-learn, scaling was only performed on the training set, not on the test set to prevent data leakage.
- **Label encoding:** the 22 crop names were encoded into integer class labels to be used with scikit-learn classifiers, with the help of LabelEncoder.

Machine Learning Architectures

Based on the same scaled feature set, five supervised classification methods were trained to be able to make meaningful comparisons of their performance: Random Forest, Gradient Boosting, Support Vector Machine (RBF kernel), K-Nearest Neighbors ($k = 5$), and multinomial Logistic Regression. The strongest baseline, Random Forest was further optimized using 5-fold random stratified cross-validation, by optimizing the number of estimators (`n_estimators`), depth of tree (`max_depth`), minimum number of samples in a split (`min_samples_split`), and minimum number of samples required to be at a leaf (`min_samples_leaf`). The best configuration factors using (`n_estimators = 200`, `max_depth = None`, `min_samples_split = 5`, `min_samples_leaf = 1`) achieved an outstanding score in cross-validated macro-F1 of 0.9954.

Full Workflow / Pipeline

- **Data Preparation**
 - a) load and validate data of the CSV file
 - b) rearranging training and test sets into stratified sampling method
 - c) scaling data
 - d) encode their target labels
- **Model Training**

Run a training for all five candidate classifiers on the scaled training data and test each of them against the held-out test data with accuracy and macro-averaged precision, recall and F1 scores.
- **Optimization & Deployment**
 1. Tune the best performing model using GridSearchCV
 2. extract feature importance
 3. save final model with scaler and label encoder (via joblib)
 4. expose a single function for making inferences that reads new readings and returns a predicted crop.

Results

The presented test set consisted of 440 samples (20 % of the total), which were used for all five candidate models to test them. Table 1 shows accuracy and macro-averaged precision, recall, and F1-scores for each model including the Random Forest that was tuned.

Table 1. Test-set performance of all trained models.

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Random Forest (baseline)	99.55%	99.57%	99.55%	99.55%
Gradient Boosting	98.86%	98.97%	98.86%	98.87%
SVM (RBF kernel)	98.41%	98.56%	98.41%	98.40%
K-Nearest Neighbors	97.95%	98.04%	97.95%	97.93%
Logistic Regression	97.27%	97.40%	97.27%	97.25%
Random Forest (tuned)	99.55%	99.57%	99.55%	99.55%

Figure 1 shows the macro-F1 comparison over all 5 baseline models. The seven features of input data were able to cleanly separate 22 different crop classes, with all five classifiers achieving macro-F1 scores greater than 97%. Amongst the baselines, Random Forest and Gradient Boosting performed excellently (99.55% and 98.87% respectively), with both being Ensemble Methods, and they are generally better at dealing with non-linear feature interactions than say a single decision boundary such as Logistic Regression which performed the lowest of the five (97.25%). Random Forest was chosen for hyperparameter tuning because the tuned version was close to the untuned version to two decimal places, indicating it was unlikely that the default values were far from optimal for this dataset because the relatively small amount of error was likely attributable to a few samples that were clearly ambiguous and not under-fitting.

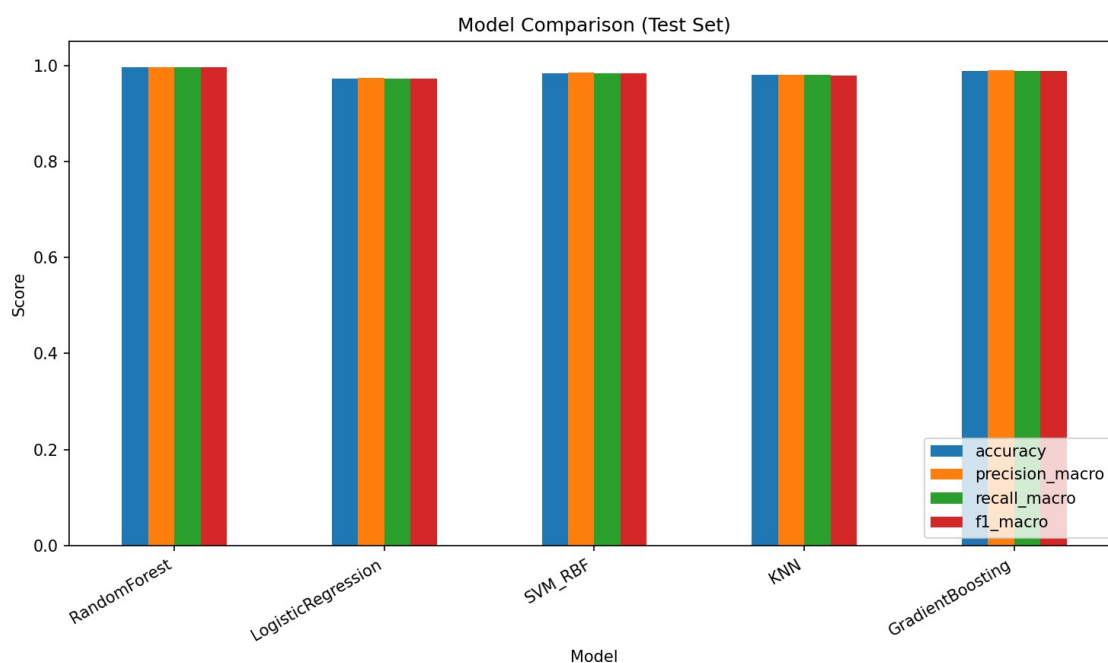


Figure 1. Macro-F1, accuracy, precision, and recall across all five models (test set).

In figure 2, The feature importance obtained from the tuned Random Forest is plotted. Together, rainfall and humidity contributed about 44% of the model's decisions, second only to the combined effects of potassium (17.8%), phosphorus (14.9%) and nitrogen (10.7%). The contribution of temperature (7.5%) and pH (5.1%) was the lowest. This suggests that climatic factors, especially the availability of water, are more important than soil nutrients in separating the crops in this dataset, a behavior consistent with the fact that many of the crops in the dataset (such as rice, jute, coconut and various other trees) also have differing demands for water and humidity.

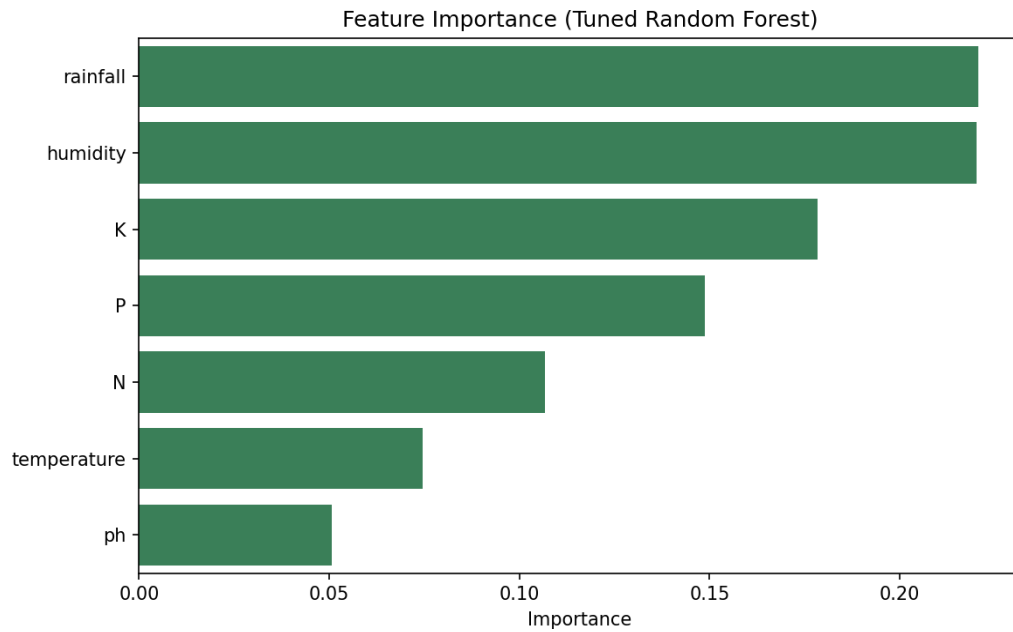


Figure 2. Ranking of Feature importance using the tuned RF model.

In Figure 3, the confusion matrix with the tuned Random Forest on test set is presented. The strong diagonal suggests that, other than a few misclassifications on very similar crop pairs visually and agronomically (e.g., moth beans versus other legumes), the misclassifications were generally clustered and not randomly distributed across all different crop pairs. The small residual error from the similar pattern indicates underperformance of model overlap in growing conditions.

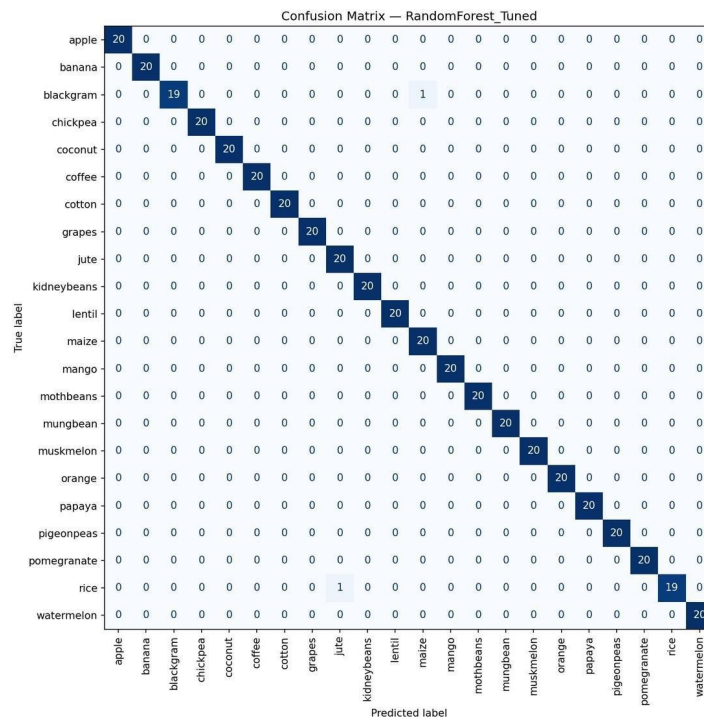


Figure 3. Confusion matrix of the tuned Random Forest on the held-out test set.

Conclusion

Tuned RF was the model that was chosen, and it achieved an accuracy of 99.55% and macro-F1 of 99.55% on an unseen test set. Ensemble tree-based method classification models (Random Forest and Gradient Boosting) are overall more successful at this task than linear classifiers and distance-based classifiers (Logistic Regression and KNN) and climatic features (rainfall and humidity) are the most predictive for crop suitability. The final model, along with its feature scaler and label encoder, were saved for reuse so that in the future the model could be used to predict the crop for any new soil and climate readings without any retraining performed.